Lecture Notes

Lecture 5

Eigenvalues and Eigenvectors

Lecture 4 ended with a tantalising example: the right basis turned a reflection into the diagonal matrix $\operatorname{diag}(1, -1)$. The diagonal entries were the factors by which the operator stretched two special directions. This lecture names those objects—*eigenvectors*, the directions an operator merely scales, and *eigenvalues*, the scaling factors—and develops the tools to find them: the characteristic polynomial, the guaranteed existence of eigenvalues over $\mathbb{C}$, and upper-triangular form, in which the eigenvalues sit on the diagonal.

Unless an example is explicitly labelled otherwise, V is a nonzero finite-dimensional vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $T \in \mathcal{L}(V)$. Thus the eigenvalue set studied here is the *point spectrum*; in this finite setting it agrees with the general spectrum defined by failure of invertibility. Lecture 14 separates point, continuous, and residual spectrum for operators on Hilbert space.

In the finite/discrete quantum model, a self-adjoint observable has eigenvalues for projective outcomes and eigenspaces for definite values. The equation $\hat{H}\psi = E\psi$ describes a discrete energy eigenstate. General observables may also have continuous spectrum, and atomic spectral lines record transition energy differences; Lectures 10 and 14 supply those qualifications.

Learning objectives

By the end of this lecture you will be able to:

- Identify invariant subspaces, eigenvalues, eigenvectors, and eigenspaces.
- Compute eigenvalues from the characteristic polynomial and recover tr and det.
- Explain why every operator on a nonzero finite-dimensional complex space has an eigenvalue (and why real failure is possible).
- Show eigenvectors for distinct eigenvalues are independent; triangularise over $\mathbb{C}$.

1 Invariant subspaces and eigenvectors

Guiding question. Which parts of a vector space keep their identity when an operator acts? Given $T \in \mathcal{L}(V)$, we look first for subspaces that T preserves.

Definition 5.1 (Invariant subspace). A subspace $U \subseteq V$ is *invariant* under T when every vector that starts in U remains in U after applying T:

$$u \in U \quad \Longrightarrow \quad Tu \in U.$$

Equivalently,

$$T(U) \subseteq U.$$

Start with familiar invariant subspaces. The following are always invariant:

$$V, \qquad \{0\}, \qquad \operatorname{null} T, \qquad \operatorname{range} T.$$

The smallest nontrivial invariant subspaces are lines. For $v \neq 0$, the line $\operatorname{span}(v)$ is invariant exactly when Tv is a scalar multiple of v.

Definition 5.2 (Eigenvalue, eigenvector, eigenspace). A nonzero vector keeps its line under T precisely when

$$Tv = \lambda v, \qquad v \neq 0.$$

- **Eigenvalue.** The scalar $\lambda \in \mathbb{F}$ is the scale factor.
- **Eigenvector.** The nonzero vector v is a direction that T preserves.
- **Eigenspace.** All vectors with the same scale factor form the subspace

  $$E(\lambda, T) = \operatorname{null}(T - \lambda I) = \{v \in V : Tv = \lambda v\},$$

  including the zero vector.

The set of eigenvalues is the *point spectrum*. In our standing finite-dimensional setting it equals $\operatorname{spec}(T)$, defined by failure of invertibility. Lecture 14 explains why this equality can fail in infinite dimension.

Three equivalent tests. The eigenspace is a kernel, so for a scalar λ we may move freely along the chain

$$\begin{aligned} \lambda \in \operatorname{spec}(T) &\iff E(\lambda, T) \neq \{0\} \\ &\iff T - \lambda I \text{ is not injective} \\ &\iff T - \lambda I \text{ is not invertible.} \end{aligned} \tag{5.1}$$

The last step uses the finite-dimensional equivalence "injective iff invertible" from Lecture 3. Geometrically, an eigenvector is a direction that T merely rescales (Figure 1).

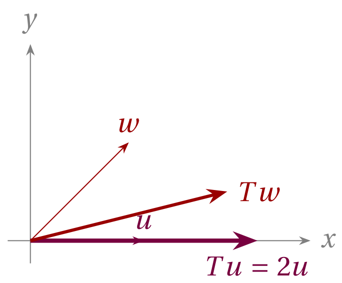

Figure 1: For the stretch $T(x, y) = (2x, \frac{1}{2}y)$, the x-axis is an eigendirection ($Tu = 2u$); a generic vector w is turned, so it is not an eigenvector.

Example 5.3 (A catalogue of eigenvalues).

- **Projection.** If $P^2 = P$, its eigenvalues lie in $\{0, 1\}$: $E(1, P) = \operatorname{range} P$ and $E(0, P) = \operatorname{null} P$.
- **Reflection.** Reflection across a line in $\mathbb{R}^2$ has eigenvalues $+1$ (along the line) and -1 (across it), as we saw in Lecture 4.
- **Differentiation.** On smooth functions, every scalar λ is an eigenvalue because

  $$De^{\lambda t} = \lambda e^{\lambda t}.$$

- **Rotation.** A rotation R_θ of $\mathbb{R}^2$ with $\theta \notin \pi\mathbb{Z}$ has *no* real eigenvalue—it turns every nonzero vector (Figure 2). ◁

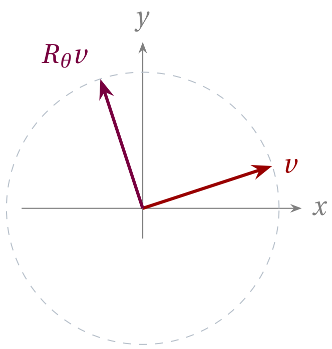

Figure 2: When $\theta \notin \pi\mathbb{Z}$, a rotation moves every nonzero real vector off its own line, so it has no real eigenvector. Over $\mathbb{C}$ its eigenvalues are $e^{\pm i\theta}$.

Example 5.4 (Computing eigenvalues of a 2×2 matrix). Find all eigenvalues and eigenspaces of the operator on $\mathbb{R}^2$ represented by

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Before reading the solution, predict the eigendirections from the symmetry of the matrix.

1. **Find the possible scale factors.** Singularity requires

$$\det(A - \lambda I) = (2-\lambda)^2 - 1 = 0,$$

   hence

$$\lambda = 3 \qquad \text{or} \qquad \lambda = 1.$$

2. **Find the direction for $\lambda = 3$.**

$$(A - 3I)v = 0 \qquad \Longleftrightarrow \qquad -v_1 + v_2 = 0,$$

   so

$$E(3, A) = \operatorname{span}\big((1,1)\big).$$

3. **Find the direction for $\lambda = 1$.**

$$(A - I)v = 0 \qquad \Longleftrightarrow \qquad v_1 + v_2 = 0,$$

   so

$$E(1, A) = \operatorname{span}\big((1,-1)\big).$$

Interpretation. The two eigendirections lie along and across the line $y = x$, just as the symmetric form of A suggests. ◁

Example 5.5 (Invariant but not an eigenline). Invariance is weaker than being spanned by an eigenvector. Consider

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

- **Eigenline.** The x-axis is invariant and its nonzero vectors have eigenvalue 1.
- **Larger example.** The whole space $\mathbb{R}^2$ is invariant, but it is not a line.
- **Non-example.** The line $\operatorname{span}\big((1,1)\big)$ is not invariant because

$$A(1,1) = (2,1) \notin \operatorname{span}\big((1,1)\big).$$

Only special lines are eigenlines. ◁

Example 5.6 (Physics bridge: the Pauli matrix σ_y). The observable

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

is genuinely complex. Its characteristic equation is

$$\det(\lambda I - \sigma_y) = \lambda^2 - 1 = 0,$$

so its two eigenpairs are

$$\lambda = +1, \quad v_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \lambda = -1, \quad v_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

Physical meaning. These are spin states pointing along $+y$ and $-y$, the spin analogue of circular polarization. Neither has a real representative: quantum state space is naturally complex. ◁

2 Computing eigenvalues and eigenspaces

Guiding question. How do we turn the geometric search for preserved directions into a reliable calculation? In coordinates, the link is singularity, detected by a determinant.

Definition 5.7 (Characteristic polynomial). For $A \in M_n(\mathbb{F})$ (or for $T \in \mathcal{L}(V)$ via any basis), the *characteristic polynomial* is

$$\chi_A(\lambda) = \det(\lambda I - A).$$

It is monic of degree n. Its roots are exactly the eigenvalues of A:

$$\chi_A(\lambda) = 0 \quad \Longleftrightarrow \quad \lambda \text{ is an eigenvalue of } A.$$

The eigenpair method. For a concrete matrix A, separate the scale-factor calculation from the direction calculation.

1. **Form the characteristic polynomial.**
$$\chi_A(\lambda) = \det(\lambda I - A).$$
2. **Find the possible scale factors.**
$$\chi_A(\lambda) = 0.$$
3. **Find the directions for each root.**
$$E(\lambda, A) = \ker(A - \lambda I).$$
4. **Check the complete result.** Compare the eigenvalues with $\operatorname{tr} A$ and $\det A$, and count the independent eigenvectors.

Do not stop at the roots. The polynomial finds possible scale factors; the kernel calculation finds the directions. Reporting roots without eigenspaces gives only half of the eigenpair calculation.

Remark 5.8 (Determinant facts we use). This course uses four determinant facts without reproving them here.

- **Invertibility.** A square matrix is invertible iff $\det A \neq 0$.
- **Products.** $\det(AB) = \det A \, \det B$. In particular,

$$\det(P^{-1}AP) = \det A.$$

- **Triangular form.** The determinant of a triangular matrix is the product of its diagonal entries.
- **Coefficients.**

$$\det(\lambda I - A) = \lambda^n - (\operatorname{tr} A)\lambda^{n-1} + \cdots + (-1)^n \det A.$$

Each follows from cofactor expansion or row reduction and is proved in the course texts.

Why the roots are eigenvalues. The key implications form one chain:

$$\chi_A(\lambda) = 0 \iff \lambda I - A \text{ is singular} \iff \ker(\lambda I - A) \neq \{0\} \iff \lambda \text{ is an eigenvalue.}$$

Why the polynomial belongs to the operator. If two matrices represent the same operator in different bases, they are similar. Then

$$\det(\lambda I - P^{-1}AP) = \det\big(P^{-1}(\lambda I - A)P\big) = \det(\lambda I - A),$$

and therefore

$$\chi_{P^{-1}AP} = \chi_A.$$

Thus the characteristic polynomial and its eigenvalues do not depend on the chosen basis.

Proposition 5.9 (Trace and determinant from eigenvalues). Over $\mathbb{C}$, if $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A (listed with multiplicity, i.e. the roots of χ_A), then

$$\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n, \qquad \det A = \lambda_1 \cdots \lambda_n.$$

Proof. Write the same monic polynomial in two ways:

$$\chi_A(\lambda) = \lambda^n - (\operatorname{tr} A)\lambda^{n-1} + \cdots + (-1)^n \det A,$$

and

$$\chi_A(\lambda) = \prod_{k=1}^{n}(\lambda - \lambda_k) = \lambda^n - \left(\sum_{k=1}^{n} \lambda_k\right)\lambda^{n-1} + \cdots + (-1)^n \prod_{k=1}^{n} \lambda_k.$$

Matching the coefficient of λ^{n-1} and the constant term gives the two identities. ■

Example 5.10 (A 2×2 characteristic polynomial). For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ of Example 5.4,

$$\chi_A(\lambda) = (\lambda - 2)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The roots recover the eigenvalues 1 and 3.

$$\underbrace{\operatorname{tr} A}_{4} = 1 + 3, \qquad \underbrace{\det A}_{3} = 1 \cdot 3.$$

Both checks pass. ◁

Example 5.11 (The 2×2 characteristic polynomial in general). For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{C}),$$

$$\chi_A(\lambda) = \det \begin{pmatrix} \lambda - a & -b \\ -c & \lambda - d \end{pmatrix} = \lambda^2 - (a+d)\lambda + (ad - bc) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A,$$

the eigenvalues are

$$\lambda_\pm = \frac{1}{2}\left(\operatorname{tr} A \pm \sqrt{(\operatorname{tr} A)^2 - 4 \det A}\right).$$

For a real matrix, the discriminant

$$\Delta = (\operatorname{tr} A)^2 - 4 \det A$$

has a geometric reading: $\Delta > 0$ gives two real eigendirections, $\Delta = 0$ a repeated real eigenvalue, and $\Delta < 0$ a nonreal complex-conjugate pair. For a general complex matrix there is no sign classification. ◁

Example 5.12 (A 3×3 characteristic polynomial). For the lower-triangular matrix

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix},$$

$$\chi_A(\lambda) = \det \begin{pmatrix} \lambda - 2 & 0 & 0 \\ -1 & \lambda - 3 & 0 \\ 0 & -1 & \lambda - 2 \end{pmatrix} = (\lambda - 2)(\lambda - 3)(\lambda - 2) = (\lambda - 2)^2(\lambda - 3),$$

so the eigenvalues are

$$\lambda = 2 \quad \text{with multiplicity } 2, \qquad \lambda = 3 \quad \text{with multiplicity } 1.$$

Here the triangular form lets us read the factors directly from the diagonal. ◁

Example 5.13 (From eigenvalues to eigenspaces). For the same $A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$, find $E(3, A)$ and $E(2, A)$. Compare each dimension with the corresponding root multiplicity in Example 5.12.

1. For $\lambda = 3$, the system $(A - 3I)v = 0$ gives

   $$v_1 = 0, \qquad v_2 = v_3,$$

   hence

   $$E(3, A) = \operatorname{span}\big((0, 1, 1)\big).$$

2. For $\lambda = 2$, the system $(A - 2I)v = 0$ gives

   $$v_1 + v_2 = 0, \qquad v_2 = 0,$$

   hence

   $$E(2, A) = \operatorname{span}\big((0, 0, 1)\big).$$

Interpretation. The eigenvalue 2 has algebraic multiplicity 2 but geometric multiplicity 1. The matrix is defective and cannot be diagonalized. ◁

Example 5.14 (Complex eigenvalues of a rotation). The rotation matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

has characteristic polynomial

$$\chi(\lambda) = (\lambda - \cos\theta)^2 + \sin^2\theta = \lambda^2 - 2\cos\theta\,\lambda + 1.$$

Its roots are

$$\lambda_\pm = \cos\theta \pm i\sin\theta = e^{\pm i\theta}.$$

There are no real roots unless $\theta \in \pi\mathbb{Z}$, confirming Figure 2. Over $\mathbb{C}$, the two eigenvalues lie on the unit circle. ◁

3 Eigenvectors for distinct eigenvalues are independent

Guiding question. When can several eigenvectors be used as independent coordinate directions? Distinct scale factors provide exactly the separation we need.

> **Theorem 5.15 (Independence of eigenvectors).** Assume
>
> $$Tv_j = \lambda_j v_j \qquad (j = 1, \ldots, m),$$
>
> where every v_j is nonzero and the eigenvalues $\lambda_1, \ldots, \lambda_m$ are distinct. Then
>
> $$v_1, \ldots, v_m \quad \text{are linearly independent.}$$

Proof. **Strategy.** Assume there is a first dependence, then use the eigenvalue equation to remove its final vector.

1. Let k be the smallest index for which

$$v_k \in \operatorname{span}(v_1, \ldots, v_{k-1}).$$

   Then $v_1, \ldots, v_{k-1}$ are independent, and for some scalars a_j,

$$v_k = \sum_{j<k} a_j v_j.$$

2. Apply T and use $Tv_j = \lambda_j v_j$:

$$\lambda_k v_k = \sum_{j<k} a_j \lambda_j v_j.$$

3. Multiply the original relation by λ_k:

$$\lambda_k v_k = \sum_{j<k} a_j \lambda_k v_j.$$

4. Subtract the two equations:

$$0 = \sum_{j<k} a_j (\lambda_j - \lambda_k) v_j.$$

5. Independence of $v_1, \dots, v_{k-1}$ gives

$$a_j(\lambda_j - \lambda_k) = 0 \qquad (j < k).$$

Distinctness makes every factor $\lambda_j - \lambda_k$ nonzero, so every $a_j = 0$. This would force $v_k = 0$, contradicting the definition of an eigenvector.

■

Why this matters. Different eigenvalues automatically produce different independent directions. No row reduction of the eigenvectors is required.

> **Corollary 5.16 (At most $\dim V$ eigenvalues).** An operator on an n-dimensional space has at most n distinct eigenvalues.

Proof. Eigenvectors for distinct eigenvalues form an independent list, which can have at most $n = \dim V$ entries. ■

> **Corollary 5.17 (Eigenspaces are independent).** For distinct eigenvalues $\lambda_1, \dots, \lambda_m$,
>
> $$E(\lambda_1, T) + \cdots + E(\lambda_m, T) = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T).$$

Proof. A nontrivial way of writing 0 as a sum $u_1 + \cdots + u_m$ with $u_j \in E(\lambda_j, T)$ and not all $u_j = 0$ would be a vanishing combination of eigenvectors for distinct eigenvalues, contradicting Theorem 5.15. Hence the only representation of 0 is trivial, so the sum is direct. ■

The bridge to diagonalization. On an n-dimensional space,

$$n \text{ distinct eigenvalues} \quad \Longrightarrow \quad n \text{ independent eigenvectors} \quad \Longrightarrow \quad \text{an eigenbasis.}$$

In that basis the matrix is diagonal. Lecture 6 develops the complete diagonalization criterion.

Example 5.18 (A diagonalizing basis). From Example 5.4,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

has eigenpairs

$$\lambda_1 = 3, \quad v_1 = (1,1), \qquad \lambda_2 = 1, \quad v_2 = (1,-1).$$

Use the eigenvectors as the columns of

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$P^{-1}AP = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \operatorname{diag}(3,1),$$

with the eigenvalues displayed on the diagonal. ◁

4 Existence of eigenvalues over $\mathbb{C}$

Guiding question. Why can a real rotation have no eigenvector while a complex operator is guaranteed to have one? The difference is whether the characteristic polynomial must have a root.

Theorem 5.19 (Eigenvalues exist over $\mathbb{C}$). If

$$V \neq \{0\}, \qquad \dim V < \infty, \qquad V \text{ is a complex vector space,}$$

then every $T \in \mathcal{L}(V)$ has an eigenvalue.

Proof. Choose a basis and let A represent T. The proof is the chain

$$\deg \chi_A = \dim V \geq 1 \xRightarrow{\text{fundamental theorem of algebra}} \chi_A(\lambda) = 0 \text{ for some } \lambda \in \mathbb{C}$$
$$\Longrightarrow \det(\lambda I - A) = 0$$
$$\Longrightarrow \ker(\lambda I - A) \neq \{0\}$$
$$\Longrightarrow Av = \lambda v \text{ for some } v \neq 0.$$

Thus λ is an eigenvalue of T. ■

Remark 5.20.
- **Where $\mathbb{C}$ enters.** The fundamental theorem of algebra supplies a root. A real polynomial need not have one.
- **What the theorem does not say.** Complex scalars also encode phase and interference, but measurement values require self-adjointness and the Born postulate; eigenvalue existence alone is not a measurement theory.

Remark 5.21 (A coordinate-free extension). The determinant-free proof has four moves.

1. Pick $v \neq 0$. If $n = \dim V$, the $n+1$ vectors
$$v, Tv, \dots, T^n v$$
are linearly dependent.
2. Therefore a nonconstant polynomial p satisfies
$$p(T)v = 0.$$
3. Over $\mathbb{C}$, factor
$$p(z) = c(z - \lambda_1)\cdots(z - \lambda_m).$$
4. If every $T - \lambda_j I$ were injective, their product could not send $v \neq 0$ to zero. At least one factor therefore has a nonzero kernel.

This elegant argument is an extension; the characteristic-polynomial proof is the core route.

Example 5.22 (The existence proof in action). Consider the quarter-turn

$$T(z_1, z_2) = (z_2, -z_1), \qquad A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Over $\mathbb{C}$,

$$\chi_A(\lambda) = \lambda^2 + 1 = (\lambda - i)(\lambda + i),$$

so

$$\operatorname{spec}(A) = \{i, -i\}.$$

Over $\mathbb{R}$, the same polynomial has no root, matching the geometry: a quarter-turn preserves no real line. ◁

Extension boundary. The remainder of this section is an optional structural consequence. It is not needed for the core Lecture 5 route.

Proposition 5.23 (Commuting operators share an eigenvector *[extension]*). If V is a nonzero finite-dimensional complex vector space and $S, T \in \mathcal{L}(V)$ commute, then S and T have a common eigenvector.

Proof.
1. Choose an eigenvalue λ of S, which exists by Theorem 5.19.
2. Its eigenspace is invariant under T. Indeed, for $v \in E(\lambda, S)$,
$$S(Tv) = T(Sv) = T(\lambda v) = \lambda(Tv).$$
Thus $Tv \in E(\lambda, S)$.
3. Restrict T to the nonzero complex space $E(\lambda, S)$. By Theorem 5.19, this restriction has an eigenvector v.
4. This v is an eigenvector of T and, because it lies in $E(\lambda, S)$, also of S.

■

Triangularisation is not diagonalisation. Repeating the invariant-subspace argument gives

$$\text{commuting complex operators} \implies \text{simultaneous upper-triangularisation.}$$

For simultaneous diagonalisation, the operators must additionally be diagonalizable. Pairwise commuting normal operators satisfy this later criterion. The identity and a non-diagonalizable Jordan block commute, but they cannot be simultaneously diagonalized.

5 Multiplicity and defective operators

Guiding question. A repeated eigenvalue appears several times in the characteristic polynomial. Does it necessarily provide the same number of independent eigenvectors?

Two multiplicities. For an eigenvalue λ, keep two counts separate:

- **Algebraic multiplicity:** the multiplicity of λ as a root of χ_T.
- **Geometric multiplicity:**
$$\dim E(\lambda, T) = \dim \ker(T - \lambda I).$$

The calculation in Example 5.13 shows that these counts can differ.

Proposition 5.24 (Geometric $\le$ algebraic). For every eigenvalue λ,
$$\dim E(\lambda, T) \le \operatorname{mult}_{\chi_T}(\lambda).$$

Proof. Let $g = \dim E(\lambda, T)$.
1. Choose a basis $v_1, \dots, v_g$ of $E(\lambda, T)$ and extend it to a basis $v_1, \dots, v_n$ of V.
2. Since $Tv_k = \lambda v_k$ for $k \le g$, the matrix has block form
$$\mathcal{M}(T) = \begin{pmatrix} \lambda I_g & B \\ 0 & C \end{pmatrix}.$$
3. Its characteristic polynomial therefore factors as
$$\chi_T(t) = (t - \lambda)^g \det\bigl(tI_{n-g} - C\bigr).$$
4. Hence $(t - \lambda)^g$ divides $\chi_T(t)$. The root multiplicity of λ is at least g.

■

When diagonalisation fails. An operator is *defective* when an eigenvalue supplies fewer independent eigenvectors than its algebraic multiplicity predicts. The standard example is the shear

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Its two counts are

$$\underbrace{\operatorname{mult}_{\chi_S}(1)}_{2} > \underbrace{\dim E(1,S)}_{1}, \qquad E(1,S) = \operatorname{span}\big((1,0)\big).$$

It therefore has too few eigenvectors to form a basis.

Example 5.25 (A full diagonalization). Diagonalize

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 1 & 0 & 5 \end{pmatrix}.$$

1. **Read the eigenvalues.** Since A is triangular,

   $$\lambda = 2, \quad 3, \quad 5.$$

   They are distinct, so A is diagonalizable.
2. **Find one vector in each eigenspace.**

   $$v_2 = (-3,0,1), \qquad v_3 = (0,1,0), \qquad v_5 = (0,0,1).$$

3. **Use the eigenvectors as columns.**

   $$P = \begin{pmatrix} -3 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \operatorname{diag}(2,3,5).$$

4. **Check.**

   $$\operatorname{tr} A = 10 = 2+3+5, \qquad \det A = 30 = 2 \cdot 3 \cdot 5.$$

Both invariants agree with Proposition 5.9. ◁

Extension boundary: generalized eigenvectors. The remaining material in this section bridges to optional Jordan theory in Lecture 6. It is not part of the core route or progress requirements, and the rest of the course does not depend on it.

Definition 5.26 (Generalized eigenvectors and eigenspaces *[self-study]*). A nonzero vector v is a *generalized eigenvector* for λ when

$$(T-\lambda I)^k v = 0 \qquad \text{for some } k \geq 1.$$

The generalized eigenspace is

$$G(\lambda,T) = \{v : (T-\lambda I)^k v = 0 \text{ for some } k\} = \ker(T-\lambda I)^{\dim V},$$

and

$$E(\lambda,T) \subseteq G(\lambda,T).$$

Theorem 5.27 (Generalized eigenspace decomposition·cited). Over $\mathbb{C}$, $\dim G(\lambda, T)$ equals the algebraic multiplicity of λ, and V is the direct sum of the generalized eigenspaces,

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_r, T).$$

Interpretation. A defective operator can lack enough ordinary eigenvectors while still having enough generalized eigenvectors to span the space. This is the structure behind Jordan form. The decomposition is cited rather than proved.

Example 5.28 (A generalized eigenvector). For the shear

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the only ordinary eigenspace is

$$E(1, T) = \operatorname{span}((1, 0)).$$

However,

$$(T - I)^2 = 0, \qquad G(1, T) = \ker(T - I)^2 = \mathbb{C}^2.$$

In particular,

$$(0, 1) \xmapsto{T-I} (1, 0) \xmapsto{T-I} 0.$$

This *Jordan chain* links a generalized eigenvector to an ordinary one. ◁

6 Upper-triangular matrices

Guiding question. What useful structure survives when an operator has too few eigenvectors for a diagonal basis? Over $\mathbb{C}$, we can still find a basis that makes its matrix upper triangular.

The invariant-flag picture. An ordered basis produces nested subspaces

$$\operatorname{span}(v_1) \subseteq \operatorname{span}(v_1, v_2) \subseteq \cdots \subseteq \operatorname{span}(v_1, \ldots, v_n) = V.$$

Upper-triangular form means that T preserves every subspace in this flag.

Proposition 5.29 (Triangular form encodes invariance). For a basis $v_1, \ldots, v_n$, the following are equivalent:

$$\begin{aligned} & \mathcal{M}_{(v_1,\ldots,v_n)}(T) \text{ is upper triangular} \\ \iff \quad & Tv_k \in \operatorname{span}(v_1, \ldots, v_k) \quad (k = 1, \ldots, n) \\ \iff \quad & \operatorname{span}(v_1, \ldots, v_k) \text{ is invariant for every } k. \end{aligned}$$

Proof. The k-th matrix column records the coefficients of Tv_k. It has zeros below row k exactly when

$$Tv_k \in \operatorname{span}(v_1, \ldots, v_k).$$

Because the spans are nested, this column condition is equivalent to invariance of every subspace in the flag. ■

Core versus extension. The triangularizability statement below and the invariant-flag criterion are core. The quotient-induction construction in its proof is an extension sketch.

Theorem 5.30 (Triangularizability over $\mathbb{C}$·sketched). For every operator T on a finite-dimensional complex vector space, there is a basis in which

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}.$$

Extension proof sketch. Induct on $n = \dim V$.

1. By Theorem 5.19, choose an eigenvector v_1. Its line $\operatorname{span}(v_1)$ is invariant.
2. Pass to the $(n-1)$-dimensional quotient $V/\operatorname{span}(v_1)$. Invariance makes the induced map
$$\bar{T}\big(v + \operatorname{span}(v_1)\big) = Tv + \operatorname{span}(v_1)$$
well-defined.
3. By induction, choose a triangularizing quotient basis
$$\bar{v}_2, \ldots, \bar{v}_n.$$
4. Lift these vectors to $v_2, \ldots, v_n \in V$. Together with v_1, they form a basis of V satisfying
$$Tv_k \in \operatorname{span}(v_1, \ldots, v_k).$$
5. Apply Proposition 5.29 to conclude that the matrix is upper triangular.

The complex field is used only to obtain an eigenvalue at each stage. The same argument works over any field on which χ_T splits. ■

Later use. The splitting-field observation reappears in the real spectral theorem of Lecture 10. The next optional lemma supports the Cayley–Hamilton proof in Lecture 6.

Lemma 5.31 (Triangular products annihilate *[Lecture 6 bridge]*). Suppose T has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to a basis $v_1, \ldots, v_n$. Then
$$(T - \lambda_1 I)(T - \lambda_2 I)\cdots(T - \lambda_n I) = 0.$$

Proof. Upper-triangularity gives the dimension-lowering move
$$(T - \lambda_k I)v_k \in \operatorname{span}(v_1, \ldots, v_{k-1}).$$

Now induct on k.

1. For $k = 1$,
$$(T - \lambda_1 I)v_1 = 0.$$
2. Assume the first $k-1$ factors annihilate $v_1, \ldots, v_{k-1}$.
3. All factors are polynomials in T, so they commute. Apply $T - \lambda_k I$ first: it sends v_k into the span already annihilated by the preceding factors.
4. At $k = n$, the full product annihilates every basis vector and is therefore the zero operator.

■

The payoff. For a triangular matrix, the spectrum can be read directly from its diagonal.

Theorem 5.32 (Eigenvalues sit on the diagonal). If T has an upper-triangular matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then

$$\operatorname{spec}(T) = \{\lambda_1, \dots, \lambda_n\}.$$

Moreover,

$$T \text{ is invertible} \quad \Longleftrightarrow \quad \lambda_k \neq 0 \text{ for every } k.$$

Proof. If A is upper triangular, then so is $\lambda I - A$. Therefore

$$\chi_A(\lambda) = \det(\lambda I - A) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n).$$

Its roots are exactly the diagonal entries. Also,

$$\det A = \lambda_1 \cdots \lambda_n,$$

which is nonzero exactly when every diagonal entry is nonzero. ∎

Example 5.33 (Reading eigenvalues off a triangle). Without expanding a determinant, read the spectrum and decide whether

$$A = \begin{pmatrix} 3 & 7 & -2 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}$$

is invertible.

1. **Read the diagonal.**
$$\lambda = 3, \quad 3, \quad -1.$$
Thus 3 has algebraic multiplicity 2, while -1 has multiplicity 1.
2. **Check invertibility.**
$$\det A = 3 \cdot 3 \cdot (-1) = -9 \neq 0.$$
Therefore A is invertible.

◁

Example 5.34 (Triangular form exposes a defective spectrum). The shear

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is already upper triangular. Hence

$$\chi_S(\lambda) = (\lambda - 1)^2, \qquad \operatorname{spec}(S) = \{1\}.$$

But

$$E(1, S) = \operatorname{span}((1, 0))$$

is only one-dimensional, so S is not diagonalizable. Triangular form still reveals its complete spectrum immediately. ◁

Physics bridge: finite and discrete observables

The remaining examples interpret the algebra in finite quantum models. They add physical meaning, not new core techniques.

Example 5.35 (The Pauli matrix σ_x). In units of $\hbar/2$, consider

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Its characteristic polynomial and normalized eigenpairs are

$$\chi(\lambda) = \lambda^2 - 1,$$

$$\lambda = +1, \quad v_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \lambda = -1, \quad v_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Physical meaning. In the finite ideal-projective model, ± 1 are the possible outcomes and the two eigenspaces are the corresponding definite-value states. The Born rule supplies probabilities; Lecture 14 replaces this finite list by a spectral measure in the general setting. ◃

Example 5.36 (Compatible and incompatible observables). Compare

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The eigenvectors of σ_z are $(1,0)$ and $(0,1)$, but neither is an eigenvector of σ_x. Direct multiplication gives

$$\sigma_x \sigma_z \neq \sigma_z \sigma_x.$$

Interpretation. Commuting diagonalizable operators share an eigenbasis. These two self-adjoint operators do not commute and share no eigenbasis, anticipating the uncertainty relation of Lecture 9. ◃

Remark 5.37 (Finite/discrete observables and energy levels).

- **Finite or discrete pure-point model.**

  $$\hat{H}\psi = E\psi$$

  identifies energy eigenvalues and stationary state vectors. The later self-adjoint spectral theorem makes distinct eigenspaces orthogonal.
- **General self-adjoint observable.** Continuous spectrum may be present and normalizable eigenvectors may be absent. Outcomes and probabilities then require a projection-valued measure and the Born rule (Lecture 14), not the fundamental theorem of algebra.

Remark 5.38. Atomic spectral lines record allowed energy differences:

$$h\nu = E_{\text{upper}} - E_{\text{lower}}.$$

Their frequencies are not individual energy eigenvalues. Selection rules, broadening, bound-state spectra, and continuous components require the more careful treatment of Lectures 13–14.

Summary of main concepts

An eigenpair separates scale from direction:

$$Tv = \lambda v, \qquad E(\lambda, T) = \operatorname{null}(T - \lambda I).$$

In finite dimension, λ is an eigenvalue exactly when $T - \lambda I$ is not invertible.

The characteristic polynomial finds the scale factors; kernel calculations find their directions. Distinct eigenvalues give independent eigenvectors, and n distinct eigenvalues on an n-dimensional space force an eigenbasis.

Over $\mathbb{C}$, the fundamental theorem of algebra guarantees at least one eigenvalue. Repeating that existence argument along invariant subspaces yields upper-triangular form, where every eigenvalue is visible on the diagonal.

- **Eigenvalue, eigenvector, eigenspace**—$Tv = \lambda v$; $E(\lambda, T) = \operatorname{null}(T - \lambda I)$; in finite dimension, λ is an eigenvalue $\iff T - \lambda I$ is not invertible.
- **Independence of eigenvectors**—distinct eigenvalues give independent eigenvectors; at most $\dim V$ of them.
- **Existence over** $\mathbb{C}$—every operator on a nonzero finite-dimensional complex space has an eigenvalue (fundamental theorem of algebra).
- **Characteristic polynomial**—$\chi_A(\lambda) = \det(\lambda I - A)$; its roots are the eigenvalues, with $\operatorname{tr} A = \sum \lambda_k$, $\det A = \prod \lambda_k$.
- **Upper-triangular form**—exists over $\mathbb{C}$; the eigenvalues are then the diagonal entries.
- **Generalized eigenvectors**—extension recognition: the generalized-eigenspace decomposition behind Jordan form (self-study; not core progress).

Exercises for this lecture are in the companion sheet `exercises-5.pdf`.